KUNCHE PRANAY CHANDRA

AI Engineer | Generative AI | Agentic AI | LLM Systems | Applied AI Engineer
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AI Engineer with 4 years of experience designing, developing, and deploying enterprise-grade Generative AI and Agentic AI solutions across Product Engineering, Retail, Consumer Goods, Education, and Human Resources. Expertise in building production-ready AI applications powered by Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), AI agents, Multi-Agent systems, Conversational AI, and Workflow Automation.
Proven ability to translate complex business requirements into secure, scalable AI solutions that improve operational efficiency, automate enterprise workflows, and enhance user experience. Passionate about AI Platform Engineering, LLM orchestration, and building reliable, production-grade intelligent systems.

TECHNICAL SKILLS

Programming Languages: Python, SQL
Generative AI: Retrieval-Augmented Generation (RAG), Large Language Models (LLMs), Transformers, AI Agents, Multi-Agent Systems, Fine-Tuning (PEFT, LoRA), Machine Learning and Deep learning (Foundational), Tensorflow and Pytorch (Foundational)
Agentic AI & Orchestration: LangGraph, LangChain, LlamaIndex, CrewAI, Agno, LangFlow
LLM Engineering: Model Context Protocol (MCP), Agent-to-Agent(A2A), Prompt Engineering, Function Calling, Tool Calling, Structured outputs, Prompt Chaining, Context Engineering
Embeddings & Retrieval: OpenAI Embeddings, Google Text Embeddings, Sentence Transformers, Semantic Chunking, Hybrid Search, Vector Search, MMR Search, Re-Ranking
AI Platforms: Azure AI Foundry, Azure OpenAI, AWS Bedrock, Google Vertex AI
Inference & Serving: OpenAI, Azure OpenAI, Google Gemini, Anthropic Claude, vLLM, Ollama, HuggingFace
Observability & Evaluation: LangSmith, LangFuse, DeepEval, Ragas, Trulens
Guardrails/AI Safety: NeMo Guardrails, GCP Model Armor and SDP, PII Detection & Redaction, Prompt Injection Defence, Jailbreak Detection, Content Moderation, Output Validation
Conversational AI: Google Dialogflow, AWS Lex, Microsoft Bot Framework, OneReach.ai
Cloud, DevOps & Version Control: Azure, GCP, AWS, Docker, Kubernetes, Git
Databases: Vector Databases, Graph Databases, PostgreSQL, MySQL, MongoDB
Backend & APIs: FastAPI, Flask, REST APIs, WebSockets, Server-Sent Events (SSE)

WORK EXPERIENCE

AI Engineer
Deloitte US-India Offices July 2025 – Present

1. **Enterprise Multi-Agent Self-Service AI Platform:**
   - Exceeded the FY26 adoption target by enabling the platform to save 20,000+ employee hours, empowering employees to resolve HR, IT, Finance, and Policy queries through AI-powered SideKick Self-Service while significantly reducing dependency on the Global Command Centre (GCC) and walk-in IT support.
   - Architected and developed an enterprise-grade multi-agent system using LangGraph, LangChain, FastAPI, Azure OpenAI, Azure OpenAI Embeddings, PostgreSQL, PGVector, SQLAlchemy, and Alembic, providing a unified conversational interface for enterprise self-service.
   - Designed an intelligent multi-agent LLM orchestration framework using LangGraph to perform intent classification, policy-based routing, Hybrid Search-based RAG retrieval, and dynamic delegation to specialized enterprise agents for personalized, context-aware responses.
   - Built a highly scalable, production-ready FastAPI backend powering real-time AI interactions with streaming responses, asynchronous processing, persistent conversation memory, PostgreSQL persistence, APScheduler workflows, Azure AI Document Intelligence-powered document ingestion, sentiment analysis, and enterprise observability using Azure Application Insights.
   - Designed and implemented an AI-powered remote remediation engine by integrating NextThink APIs and AutoHotkey automation, enabling secure approval-based resolution of Outlook, Microsoft Teams, Druva, VPN, and endpoint-related issues without IT support intervention.
   - Integrated enterprise services including ServiceNow, Azure AI Document Intelligence, Azure Blob Storage, Azure Key Vault, Microsoft Identity, and REST APIs to build secure, scalable, production-ready AI workflows.

2. **Enterprise AI Platform & Infrastructure:**
    - Engineered a reusable Model Context Protocol (MCP) platform by exposing Google Vertex AI RAG Engine corpora as enterprise knowledge tools, deploying MCP servers on Azure App Service, and enabling multiple internal AI applications to dynamically discover and invoke enterprise knowledge through standardized MCP interfaces.
    - Designed an end-to-end AI evaluation and observability framework using DeepEval, Ragas, LangSmith, and Azure Application Insights, enabling automated pre-deployment benchmarking, post-deployment monitoring, end-to-end LLM tracing, and continuous evaluation of retrieval quality, answer correctness, faithfulness, contextual relevance, and hallucination detection.
    - Architected enterprise AI security guardrails using Google Cloud Model Armor and Sensitive Data Protection APIs, implementing prompt injection protection, jailbreak detection, and automated PII identification and redaction to deliver secure, compliant, production-ready LLM applications.
    - Containerized AI applications using Docker and implemented CI/CD pipelines with Azure DevOps to automate the build, testing, and deployment of FastAPI services, LangGraph applications, and MCP servers on Azure App Service.

Gen AI Engineer
Cognizant Technology Solutions Feb 2023 – June 2025

1. **AI-Powered Voice Ordering System for PepsiCo:**
    - Developed an advanced voice-based ordering system using AWS Lex, integrated with Generative AI to enhance user experience. This AI-powered voice bot enables seamless online order placement, account updates, and order tracking through natural voice interactions. The system is designed with intent-driven workflows and dynamic slot management, leveraging AWS Lambda for backend automation, ensuring real-time processing and accuracy.

2. **LLM-Powered Chatbot with RAG Architecture:**
    - Built a sophisticated Generative AI chatbot utilizing a Large Language Model (LLM) and Retrieval-Augmented Generation (RAG) architecture. The solution processes diverse document formats using chunking and embedding techniques, storing them in a vector database for efficient retrieval. The chatbot provides accurate, context-aware responses through advanced prompt engineering and is seamlessly deployed via a Python API to support various user queries.

3. **Agentic AI-Driven Inventory and Vendor Optimization System**
    - Built an Agentic AI project that automates inventory management and vendor optimization through a multi-agent system. One agent analyzes forecasted demand and current stock levels to recommend optimal inventory decisions for upcoming market needs. Another agent dynamically selects vendors by updating weights based on real-time data like pricing, quality, and delivery performance. Additionally, a natural language agent powered by Mind AI enables querying relational databases using simple text inputs, enhancing accessibility, decision-making efficiency.
4. Developed multiple AI solutions using Azure OpenAI Studio, AWS Bedrock, and GCP AI Studio, leveraging their capabilities to build, fine-tune, and deploy Generative AI models for tasks such as text generation, summarization, sentiment analysis, and conversational AI. Integrated these models into applications to enhance automation, improve user experience, and optimize business processes across different domains.
5. Developed end-to-end Generative AI-based chatbots and applications using platforms like Google Dialogflow, AWS Lex, Microsoft Bot Framework LangChain and Onereach.ai
6. Designed and deployed both chatbots and voice bots, enhancing customer interaction and reducing human intervention.
7. Developed multiple MVP's and production ready RAG POC's using various AI frameworks and platforms, showcasing the application of Generative AI in real-world use cases.
8. Leveraged Google Cloud, AWS, and Microsoft Azure for development, deployment, ensuring scalable and secure data management across multiple cloud environments.

Data Science Intern
Cognizant Technology Solutions Feb 2022 – Aug 2022

1. Contributed to the development and maintenance of data pipelines, enabling smooth data flow from various sources for AI-driven projects.
2. Played a key role in data integration, ETL (Extract, Transform, Load) processes, and data quality assurance, ensuring accurate and reliable data for AI models.
3. Assisted in deploying and managing AI applications on Google Cloud, and Microsoft Azure, focusing on scalable cloud infrastructure and efficient data management.
4. Collaborated with cross-functional teams to streamline AI workflows and deliver cloud-based AI solutions.

PERSONAL PROJECTS

1. **AI-Powered Financial Assistant:**
   - Developed an AI-powered financial assistant using OpenAI GPT, Agno (formerly Phidata), FastAPI, and Streamlit to provide context-aware responses for financial queries through an interactive conversational interface.
   - Implemented tool-calling and agent workflows to retrieve, process, and summarize financial information, enabling intelligent query handling and dynamic response generation.
   - Designed and deployed a responsive Streamlit-based UI, providing an intuitive user experience with real-time AI-powered financial insights.
   - Deployment link: App

2. **LLM-Powered Text-to-SQL Assistant (Text2SQL):**
   - Developed an LLM-powered Text-to-SQL application using LangChain, OpenAI GPT, FastAPI, and PostgreSQL, enabling users to convert natural language questions into executable SQL queries.
   - Implemented schema-aware prompt engineering and dynamic database schema retrieval to improve SQL generation accuracy across multiple relational tables.
   - Integrated SQL query validation, execution, and error handling to generate reliable results while preventing invalid or malformed SQL statements.
   - Designed an intuitive conversational interface that allowed non-technical users to interact with databases using natural language, improving accessibility to structured data.

CERTIFICATIONS

- Microsoft Certified: Azure AI Engineer Associate (AI-102)
- Microsoft Certified: Azure AI Fundamentals (AI-900)
- Oracle Generative AI Certified
- Generative AI with Large Language models

EDUCATION

CMR Technical Campus — Hyderabad, Telangana
Bachelor of Technology — 2019 - 2022

SRRS Government Polytechnic College — Sircilla, Telangana
Diploma — 2016-2019

ACCOMPLISHMENTS

- Recognized with an Appreciation Certificate for leading Deloitte Impact Day initiatives as Cluster Lead and driving successful community engagement.
- Honored with Social Impact Award by CEO of Cognizant
- Honored with Impact Champion Award, Inspiring Volunteer Award and Change Maker Award for contribution in Skilling and Education programs – Cognizant
- Received the Young Achiever Award as Outstanding Alumni from CMR Technical Campus
- Participated In Entrepreneurship Awareness Camp conducted by JNTUH University Hyderabad
- All-round Kabaddi Player at District Level